

D.K.T.E. Society's Textile and Engineering Institute, Ichalkaranji
(An Autonomous Institute, Affiliated to Shivaji University, Kolhapur)

Accredited with 'A+' Grade by NAAC



Department of Computer Science and Engineering
(Artificial Intelligence and Machine Learning) 2026-2027.

THE COMMUNITY ENGAGEMENT PROJECT REPORT ON
“CYBER SECURITY AWARENESS SYSTEM”

Under the guidance of Prof. Rekha Kamble

Developed By:

- | | |
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CERTIFICATE

This is to certify that,

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Have successfully completed the project work, entitled

“CYBER SECURITY AWARENESS SYSTEM”

In partial fulfilment for the award of degree of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence and Machine Learning). This is record of their work carried out during academic year 2026- 27.

Prof. Rekha Kamble
[Project Guide]

Prof. Dr. S.K.Shirgave
[Head of Department]

Date: 05-10-2026

Place: Ichalkaranji

DECLARATION

We the undersigned students of S.Y.C.S.E.(AIML) declare that the Community Engagement Project work report entitled “CYBER SECURITY AWARENESS SYSTEM” written and submitted under the guidance of Prof. Rekha Kamble is our original work. The empirical findings in this report are based on the data collected by us. The matter assimilated in this report is not reproduction from any readymade report.

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INTRODUCTION

In today's fast-paced digital world, engaging in online banking and daily financial transactions has become second nature, yet it comes with significant security risks. Many individuals and small businesses face growing threats from sophisticated cyber frauds, deceptive phishing tactics, and digital identity theft. The **Cybersecurity Awareness System** is designed to address this pressing issue by bridging the gap between complex digital security practices and everyday users, providing a reliable, interactive, and user-friendly platform that empowers individuals to navigate the digital financial space safely.

The primary goal of the Cybersecurity Awareness System is to make cyber threat education simple, accessible, and practical. The application offers tailored microlearning modules designed specifically for vulnerable target groups—such as students, senior citizens, and local shopkeepers—to address their unique digital habits and risk exposures. Users can learn how to identify fake payment links, recognize fraudulent UPI alerts, and protect confidential credentials through real-time threat analyzers and interactive fraud simulators. This practical exposure ensures that users can spot deceptive patterns early and conduct online transactions with complete confidence.

For non-tech-savvy individuals and local business owners, the system offers intuitive, multi-lingual, and voice-guided interfaces that demystify cybersecurity. Rather than relying on dense technical jargon, the platform provides actionable steps on how to safeguard One-Time Passwords (OTPs), create resilient passwords, and avoid common social engineering scams.

The system also plays a crucial role in post-incident response and digital empowerment. Many victims of online financial fraud fail to recover their losses simply because they do not know where or how to report cybercrimes. By integrating an automated, one-touch incident reporting workflow connected directly to national helplines like 1930, the platform simplifies emergency reporting and accelerates financial recovery efforts.

In conclusion, the Cybersecurity Awareness System is more than just an educational tool; it is a comprehensive defense solution that addresses the practical challenges of digital fraud while fostering vigilance, confidence, and community safety. Its user-friendly design, real-time threat detection features, and focus on accessible learning make it an indispensable asset for modern digital citizens, ultimately contributing to a more secure, resilient, and digitally empowered society.

PROBLEM STATEMENT

Navigating the digital financial landscape has become increasingly risky due to the alarming rise in cyber frauds, deceptive banking scams, and online transaction exploits. Everyday users face immense difficulties in identifying manipulative tactics like fake UPI payment alerts, phishing links, and fraudulent phone calls. Traditional security awareness approaches—such as dense text guides or complex technical manuals—are inefficient, difficult to understand, and fail to reach vulnerable groups who need them most.

At the same time, specific demographics like students, senior citizens, and local shopkeepers struggle with a lack of tailored guidance and digital literacy. Unaware of basic security protocols, individuals routinely expose confidential credentials like One-Time Passwords (OTPs), PINs, and personal banking details to malicious actors. Furthermore, when cyber fraud occurs, a vast majority of victims do not know where or how to report incidents, leading to lost time, unrecovered funds, and persistent financial anxiety.

The Cybersecurity Awareness System addresses these issues by providing a unified digital platform that delivers persona-based micro-learning, interactive fraud simulations, and real-time threat analysis. By integrating a one-touch incident reporting hub connected directly to national helplines like 1930, the platform makes cyber safety intuitive, accessible, and actionable—empowering citizens to protect their assets and navigate the digital world with confidence.

PROBLEM DESCRIPTION

In the modern world, individuals and small businesses increasingly rely on online banking, UPI transfers, and digital payment systems for daily tasks and commercial transactions. However, protecting financial assets and personal identity remains a significant challenge. Most people rely on passive warnings, complex bank notices, or word-of-mouth safety tips, which are often confusing, ignored, and fail to provide practical protection against evolving fraud techniques.

Similarly, specific vulnerable groups—such as students, senior citizens, and local shopkeepers—face immense difficulties in navigating digital security. Many lack dedicated guidance tailored to their specific activities, such as online gaming scams for students, pension and call impersonation fraud for senior citizens, or fake UPI payment alerts and deceptive QR codes for shopkeepers. This absence of targeted awareness leads to widespread mistakes, with users unknowingly sharing critical credentials like One-Time Passwords (OTPs), PINs, and personal banking details with cybercriminals.

The absence of an accessible cybersecurity defense platform leads to severe financial losses, loss of trust in digital payments, and intense anxiety for victims who do not know where or how to report incidents. Most fraud victims struggle with delayed responses and complicated administrative steps, losing critical time during which stolen funds could otherwise be frozen or recovered through emergency channels.

The **Cybersecurity Awareness System** addresses these problems by providing a centralized digital platform where users can access persona-based safety training, simulate real-world cyber threats safely, and analyze suspicious text messages or links in real time. The platform features intuitive multi-lingual voice guidance for non-tech-savvy users alongside an automated, one-touch incident reporting workflow connected to national helplines like 1930. This solution not only prevents financial fraud and credential theft but also builds long-term digital confidence, creating a safer, more resilient, and well-protected online community.

FACILITY REQUIREMENTS RELATED TO THE PROBLEM

1. **Persona-Based User Dashboard:** Provide an intuitive, role-specific interface tailored for students, senior citizens, and shopkeepers.
2. **Role & Profile Management:** Allow users to set up profiles, select their demographic category, and track their personal learning progress, safety scores, and badge achievements.
3. **Interactive Micro-Learning Engine:** Offer structured, short-form modules covering topics like fake UPI alerts, phishing, social engineering, and OTP protection.
4. **AI-Driven Threat Simulator:** Enable interactive sandbox testing where users practice identifying fake SMS messages, deceptive phone calls, and phishing links safely.
5. **Real-Time Threat Analyzer:** Allow users to paste suspicious text, links, or upload QR code images to check their safety and risk scores instantly.
6. **Multi-Lingual & Voice Navigation:** Support regional languages and text-to-speech voiceovers to make content easily accessible for non-tech-savvy users and senior citizens.
7. **One-Touch Incident Reporting Hub:** Provide direct access to emergency cybercrime helplines (such as 1930) alongside an automated step-by-step reporting wizard to document fraud details.
8. **Admin Control Panel:** Enable administrators to manage learning content, update scam trend databases, monitor platform analytics, and resolve user inquiries.
9. **Centralized Database System:** Secure storage for user learning logs, threat verification histories, community reporting feeds, and regional helpline directories.

FIELD VISIT PHOTOS



Image 1: Visit to Police Station for collecting information about cyber crimes and cyber safety measures



Image 2: Visit to Court to understand the legal aspects of cyber crime



Image 3: Visit to Bank to understand banking-related cyber frauds and security measures



Image 4: Cyber Security Awareness Activity Conducted Among Students



Image 5: Cyber Security Awareness Activity Conducted Among Senior Citizens

TECHNICAL SOLUTIONS ON PROBLEM

To address digital banking fraud, targeted knowledge gaps, credential leaks, and reporting confusion, our Cybersecurity Awareness System implements key technical solutions designed to deliver a proactive, secure, and user-centered platform:

1. AI-Driven Interactive Fraud Simulator & Sandbox

- **Description:** A mobile/web module that generates simulated phishing SMS, deceptive phone calls, and fake payment links based on real-world bank fraud tactics. Users navigate these realistic scenarios in a zero-risk sandbox environment.
- **Benefits:** Teaches users how to spot subtle red flags through practice, directly reducing the likelihood of users sharing OTPs, passwords, or clicking malicious links.

2. Regional Multi-Lingual & Voice-Guided Learning Portal

- **Description:** A simplified web and app interface delivering short, interactive microcourses with voiceovers in local languages, customized for specific personas:
 - Students: Gaming scams, social media account hijacking, fake job offers.
 - Senior Citizens: Impersonation scams, pension/banking updates, OTP safety.
 - Shopkeepers: Fake UPI payment soundbox alerts, fraudulent QR codes, fake refund requests.
- **Benefits:** Removes digital and language barriers for non-tech-savvy users, ensuring critical security concepts are understood regardless of tech literacy.

3. On-Device SMS & URL Threat Analyzer

- **Description:** A privacy-focused utility built into the app where users can paste suspicious messages, links, or upload screenshots of QR codes to be analyzed in real-time using Natural Language Processing (NLP) and domain reputation engines.

- **Benefits:** Acts as a direct line of defense, warning users *before* they open dangerous links or share sensitive credentials.

4. One-Touch Incident Reporting & Helpline Integrator

- **Description:** An automated reporting wizard that features direct speed-dial access to national cybercrime helplines (such as 1930) and walks users through a quick step-by-step form to collect evidence (transaction IDs, phone numbers, screenshots) and draft official complaints.
- **Benefits:** Eliminates confusion on where and how to report cybercrimes, enabling faster action to block compromised accounts and freeze fraudulent fund transfers.

5. Gamified Community Awareness & Verification Platform

- **Description:** A community-driven platform featuring leaderboard rewards for completing safety modules, alongside a crowd-sourced repository where users can cross-check or flag newly reported scam phone numbers, fake bank UPI IDs, and fraudulent websites.
- **Benefits:** Drives user engagement through gamification while creating a real-time, user-verified database of active local scams.

AUTHORITY LETTER

Date : 28/8/2026

TO,

Shivaji Nagar Police Station (Cyber Security)
Ichalkranji, Maharashtra,
416115

Subject: visit to Shivaji Nagar Police station.
(Cyber Department) under
'Digital Literacy: Cyber Security Awareness^{-SS}
System'.

We are students of DKTE's Textile And
Engineering Institute visiting Police station
for social & cultural Activities.

We visiting for Cyber Awareness to a
Police station. This project is Helpful for
rising awareness of cyber crime.

Key components

- Community event
- Interactive Session with Experts.
- Participation & Awareness Socially.

~~28/08/2028~~

H. H. P. S. S. R.

Police Station, Ichalkranji

CONCLUSION

Cyber Security and Awareness is an essential aspect of protecting individuals, organizations, and digital information from various cyber threats. By understanding common threats such as phishing, malware, password attacks, social engineering, and data breaches, users can take appropriate measures to protect themselves and their valuable information. The implementation of strong passwords, multi-factor authentication, secure browsing practices, regular software updates, and safe handling of emails and links significantly improves digital security.

The project highlights that cyber security is not only a technical responsibility but also a shared responsibility of every digital user. Creating awareness and developing safe online habits can reduce the risk of cyber attacks and data loss. Overall, cyber security awareness promotes responsible digital behaviour, strengthens trust in digital systems, and helps create a safer and more secure digital environment for everyone.

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